HDAC-50CV
USER MANUAL

Congratulation on the purchase of the HDAC-50CV

Refrigerant Processor

This is a very good investment you have made for your workshop. This system will help your technicians in achieving their best capability in diagnosing and rectifying air-conditioning issues, thus increasing productivity and profitability for your business. This system can be customized to suit each individual workshop needs. Our company would like to work with you as partners in your business to help you achieve maximum return on your investment, so please do not hesitate to contact us or our distributors, if we can be of any further assistance in relation to this equipment use and maintenance, or any air-conditioning related issues. We are always here to help you in achieving higher productivity.

HDAC-50CV is an ideal option for workshops that have limited equipment investment budget. Even economic, the machine is robustly built, with luxurious features such as color touch screen, amply database and patented oil separator/manifold. The unique design has facilitated quick load cell unlock, easy and economic maintenance (DIY maintenance recommended), self troubleshooting and convenient USB upgrade etc..

General safety

- The storage cylinder in this unit contains liquid refrigerant. Overfilling of the cylinder can cause violent explosion. Do not disable the overfill safety feature. Always keep the cylinder on the load cell platform whenever operating the machine.
- The operator must carefully read the instruction manual before any operation is performed. Incorrect operations could cause serious consequences, such as, improper A/C service, damage to automotive A/C system or damage to equipment.
- Only use cylinders which are recommended by the manufacturer and supplied with this equipment.
- Avoid inhalation of refrigerant or oil vapor/mist, read material safety instructions on refrigerant and oil package.
- Switch off and disconnect power cable from main supply before removing any cover or servicing the equipment, to avoid electric shock which can be very dangerous or fatal.
- Never use compressed air for leak testing the unit or vehicle A/C system!
- Wear safety goggles and gloves, to protect eyes and skin from contact with refrigerant. Coming in contact with liquid refrigerant can cause frostbite and blindness. If accidental contact is made with liquid refrigerant, wash effected area with plenty of fresh water and contact a doctor.
- Avoid using extension power cable than 1.5mm^2 copper core.
- Keep gasoline or other flammable substances away from the equipment.
- Always operate unit in a well-ventilated area and away from artificial heat

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Specifications

- Input power: AC110V 60Hz
- Recovery rate: ≥95%
- Compressor output: 3/4HP(110V)
- Vacuum pump capacity: 8.0cfm
- Accuracy of gas cylinder load cell: ±30g
- Gas cylinder: 50L, 50Kg Max
- Oil bottle capacity: 400ml
- Max.Pressure: 20bar
- Charge speed: 2Kg/Min(max.)
- 7" Touch Screen display
- Refrigerant hose: 5m
- High pressure gauge range: -1~30bar
- Low pressure gauge range: -1~16bar
- Average gas state refrigerant recovery speed: 980g/min

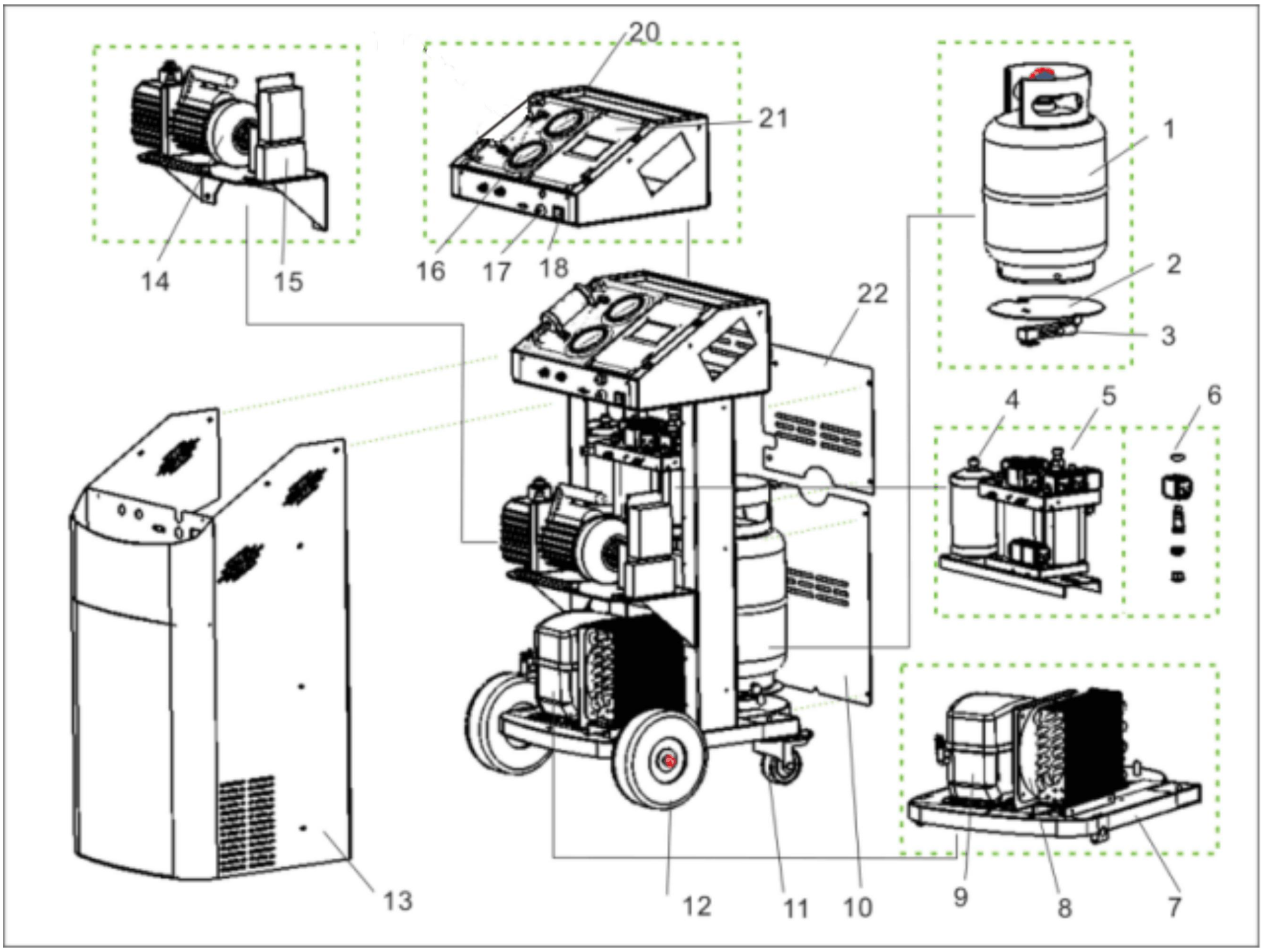
Feature

- Hand valves free
- Heater band
- Highly integrated pipeline manifold
- Condenser and cooling fan included
- Pressurization to speed up used oil discharge
- Automatic service reminding

Function Table

Main function	Recovery	Recovers and purifies refrigerant from automotive A/C to equipment tank.
	Vacuum	Evacuates air and moisture from the A/C system. Upon completion of vacuum, the machine prompts to inject new oil into A/C while a hand valve.
	Charge	Charge refrigerant from equipment gas cylinder to automotive A/C system
	Auto. mode	Performs the selected functions in a fully automatic sequence. The machine will stop automatically once all the selected functions have been completed
Sys. setting	Language	Select operation language.
	Calibration	Calibration refrigerant gas cylinder load cells.
	Database	Enter automotive A/C database
	Unit set	Select metric or imperial units
	Empty container weight set	Set empty refrigerant gas cylinder.
	Component test	Test work status of solenoids, vacuum pump and compressor.

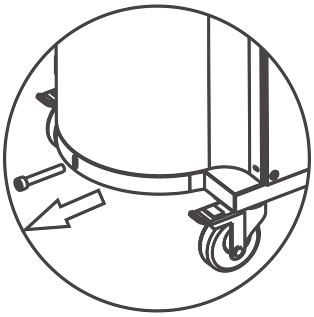
Part description



1) Refrigerant gas cylinder	2) Refrigerant gas cylinder support plate	3) Refrigerant gas cylinder load cell
4) Drier-filter	5) Manifold assembly	6) Assembly of solenoid valve and check valve
7) Base	8) Condenser and cooling fan	9) Compressor
10) Back cover1	11) Front wheel	12) Rear wheel
13) Shell	14) Vacuum pump	15) Capacitor
16) Low pressure gauge	17) Fuse	18) Power supply
19) High pressure gauge	20) Panel	21) Back cover2

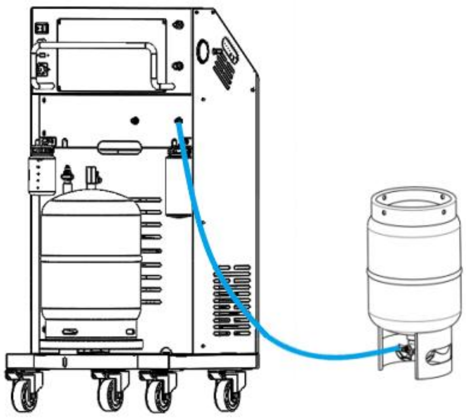
Operation preparations

■ **Unlock load cells**.



Remove the bolt that fix tank platform with chassis, to release tank load cell.

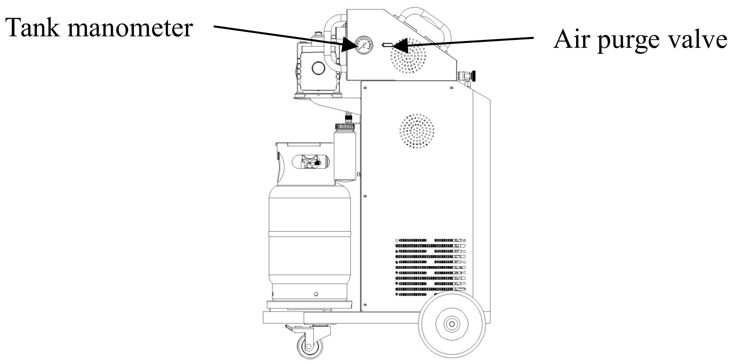
■ **Fill equipment with refrigerant** (New equipment is empty, you need to fill the equipment with refrigerant and refrigeration oil)



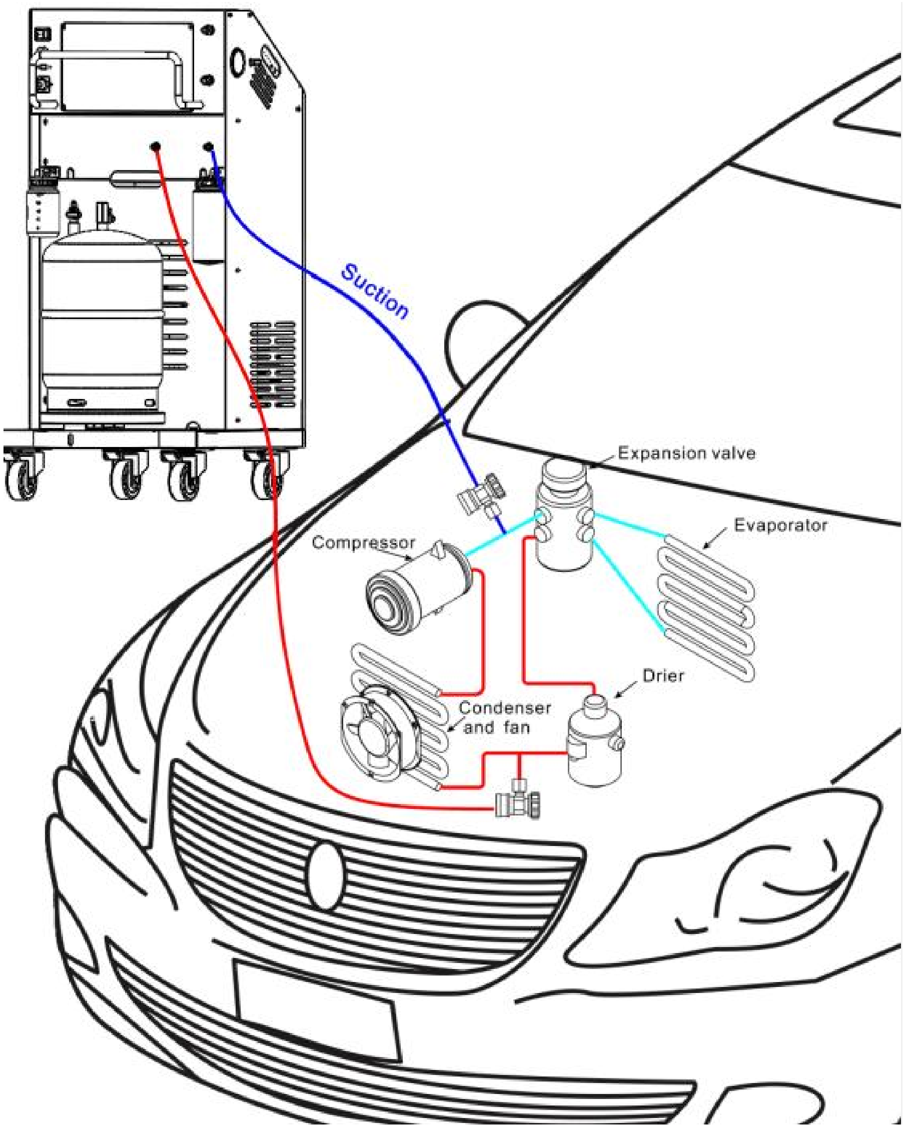
Connect either HP or LP hose with external refrigerant R134a or HFO-1234yf tank, and turn on the machine, select corresponding refrigerant type, run recovery to fill equipment with refrigerant. After one refrigerant tank is filled, fill other tank with its corresponding refrigerant.
It is recommended to maintain both tanks at 3-6kg refrigerant.

■ **Air purge**

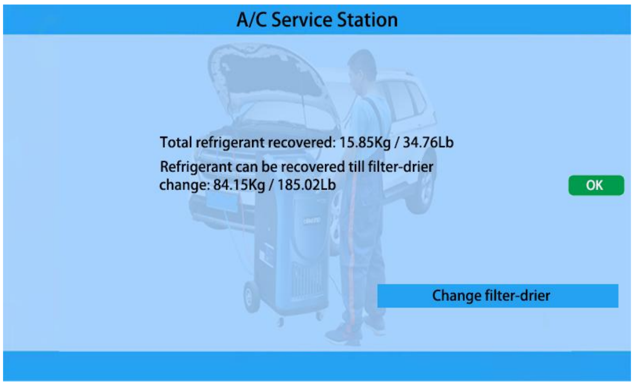
It is recommended to purge the air in equipment tanks every day before turn on the machine. At left and right sides of machine, air purge manual valves and P-T chart can be found. Turn the valve to purge the air, strictly abiding by the pressure to be consulted in chart, corresponding to environmental temperature.



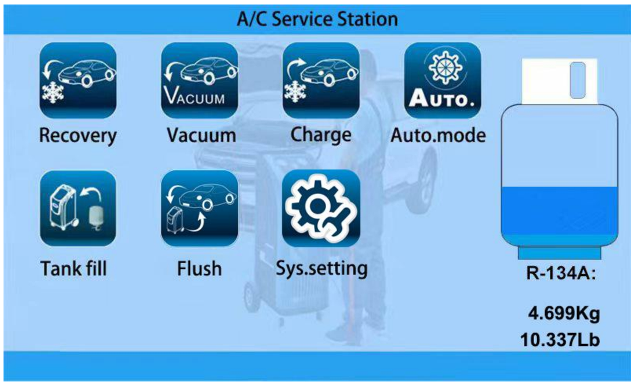
Equipment Connection



Warning: Except the situations clearly stated in the manual, during recovery and all other equipment operations, please maintain the vehicle engine and A/C off, otherwise unexpected damages may be caused.



When machine is switched on, filter-drier life is displayed. Click OK to enter main menu.



Main menu.

Recovery

Empty used oil vessel before the recovery function is started.
Select "Recovery" function icon in the main menu and click OK to start the process.

The recovery process recovers the refrigerant from vehicle A/C system, until a vacuum is achieved in the vehicle A/C system. Moisture, oil and foreign particles are separated from the refrigerant before it is stored in the internal refrigerant cylinder.

Vacuum (Vacuum and manual oil injection)

Select "Vacuum" icon in the main menu, set vacuum time and click OK to start the process.

The Vacuum process evacuates system, and makes system ready for oil injection and refrigerant charge. Although it is up to users to determine vacuum time, a longer vacuum process is recommended.

After vacuum, the machine prompts to inject oil by turning the new oil vessel hand valve at the machine side for fuel vehicles; For hybrid and electric vehicles, oil injection through the machine is prohibited, and it is recommended to inject oil with specialized oil injection tool.

Warning: PAG oil for fuel vehicle is electricity-conductive. Very few PAG oil being injected into hybrid/electric vehicle could bring about serious consequence.

Charge

Select "charge" icon and click OK to start the process.

According the vehicle being serviced, select "Normal charging" (Fuel vehicles) or "High voltage charging" (Hybrid or electric vehicles). In normal charging mode, hose flush (flush with liquid refrigerant to remove the oil residue inside the service hoses) is optional; while in high voltage charging mode, hose flush before oil injection and refrigerant charging is a must.

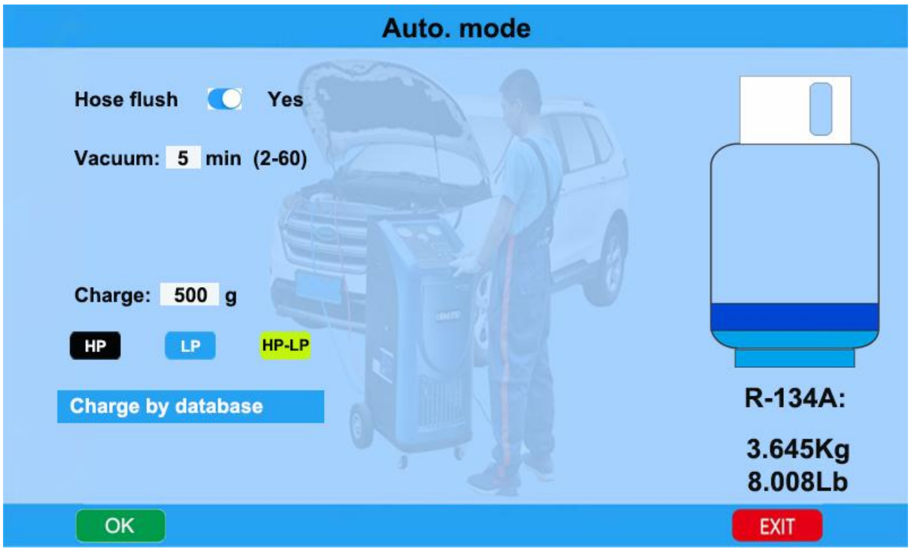
You can manually set charge amount with volume, or select "Charge by database" to set charge amount by car make and model.

You can choose to charge through high side, low side or both sides.

After charge and A/C performance is checked with engine started and A/C turned on. "Hose purge" is performed to help charge the refrigerant in service hoses into vehicle A/C system, to ensure better charge precision.

Auto. mode

Auto. Mode interface:



You can select "Auto.mode" to do full cycle of recovery, vacuum, oil injection and charge.

According the vehicle being serviced, select "Normal charging" (Fuel vehicles) or "High voltage charging" (Hybrid or electric vehicles). In normal charging mode, hose flush (flush with liquid refrigerant to remove the oil residue inside the service hoses) is optional; while in high voltage charging mode, hose flush before oil injection and refrigerant charging is a must.

In Auto. mode, the machine makes recovery, vacuum, oil injection (for fuel vehicles) and refrigerant charge in sequence automatically, with data preset by users.

Empty used oil vessel before the process.

System setting

Select “system setting” icon and input PW 111111 (left and right arrow to move cursor, up and down arrow to increase/decrease number) to enter system setting menu. In system setting, “Language”, “Calibration”, “Database”, “Unit set”, “Empty container weight set” and “Component test” can be inquired or reconfigured.

Language

Can change operation system language.

Calibration

It is suggested to have only professional technicians to do calibration of load cells. The load cell calibration is very simple and fast, with just one step, 1kg weight calibration step.

Warning: Misoperation in calibration could bring about serious consequences to equipment or vehicle A/C system.

Database

Users can access database of refrigerant/oil volume of different car makes and models.

Unit set

To set metric or American imperial unit. The two numbers displayed in the bottom part of the interface are values of two tank load cells for reference.

Empty container weight set

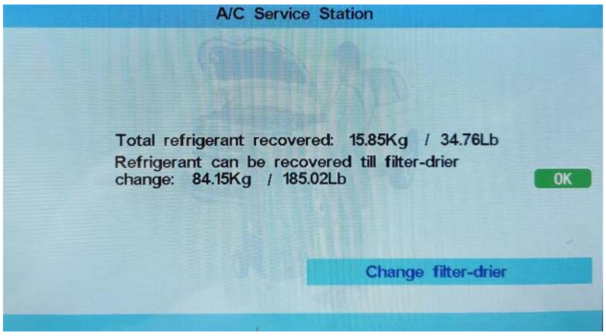
The total load cell reading equals the sum of empty container weight and net refrigerant content value. Thus, increase/decrease empty container weight, can correspondingly decrease/increase refrigerant value displayed in the main operation interface.

Component test

Users can activate/des-activate different electronic component of the machine. This is for quick and easy diagnosis for troubleshooting.

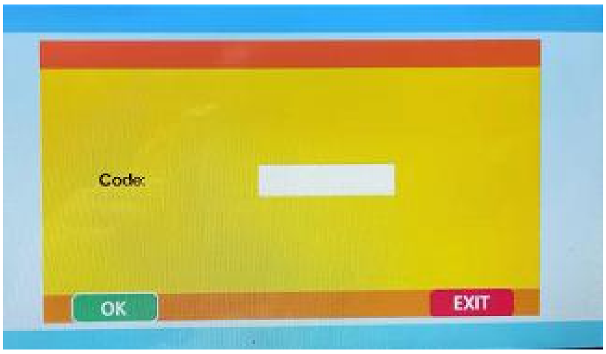
Service reminding & reset

Before total 100KG refrigerant is recovered, please select “Change filter-drier in when machine is switched on.



When this interface appears, call your equipment supplier, and send the number in the interface to get a dynamic password (each time the password is different).

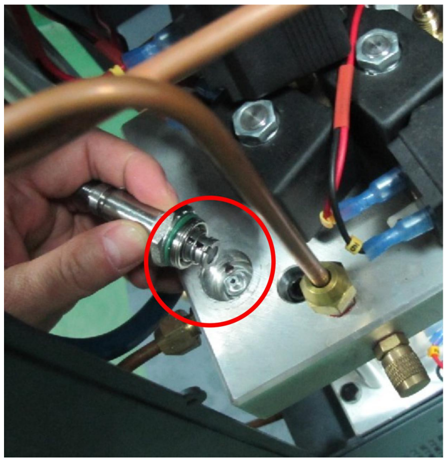
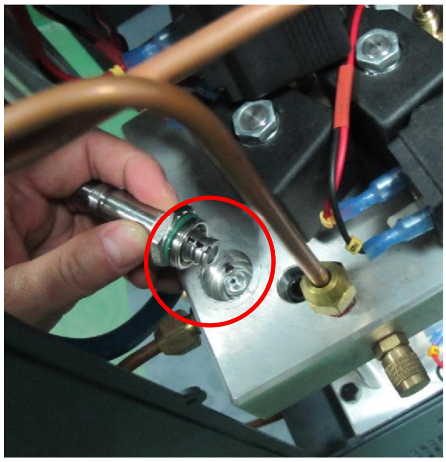
After the dynamic password is correctly input and OK is clicked. The machine asks to input filer-drier code. Please enter the code on the new filter-drier provided by the distributor.



Click OK, if the code is recognized by the machine, the machine will evacuate 60 seconds and another 100KG refrigerant is permitted to be recovered.

Main troubleshooting

Malfunction	Reasons	Solution
Low vacuum degree	1. Insufficient vacuum pump oil. 2. Pump oil emulsion, dirty 3. Pump oil inlet plugged. 4. Leakage in pump connection. 5. Components worn out.	1. Add oil to central line 2. Put new oil 3. Clean oil inlet. 4. Check connection 5. Maintain the machine, especially o-ring, washer and other sealing parts.
Vacuum pump inject oil.	1. Excessive oil volume. 2. Entrance pressure too high.	1.Discharge some oil to proper level 2.Run Recovery function first.
No display	1. Fused (in Power cable connection box, or PCA) 2. PCA burnt. 3. Power cable loosened. 4. LCD not work	1. Change fuses. 2. Change PCA. 3. Connect power cable reliably. 4. Change LCD.
Recovery does not stop	1. Leakage in automotive A/C or equipment pipeline. 2. Compressor not work 3.Pressure sensor does not work ***Remarks: In winter, it is normal that recovery takes longer time.***	1. Make leakage test. Machine leakage test with reference to service manual. 2. Change compressor. 3. Fasten pressure sensor connection to PCA, or change the pressure sensor.
No change in recovery volume	1. No refrigerant in A/C. 2. Support screw of gas cylinder load cell not loosened. 3.Gas cylinder load cell not work or PCA failure	1. Stop recovery. 2. Unscrew the protection screw, as chapter "Operation preparations". 3. Calibrate gas cylinder load cell, or change the load cell, or change PCA.
While auto A/C has refrigerant, equipment displays alarm 005	Low pressure switch plug disconnected from PCA socket.	Fasten low pressure switch plug.
High pressure alarm 004 but gas cylinder gauge does not show excessive pressure value	1. High pressure switch plug disconnected from PCA socket. 2. Pipeline connecting compressor exit blocked.	1. Fasten high pressure switch plug. 2. Inspect the hoses and connections between compressor exit and tank blue hand vale.
No charge or slow charge.	1.Insufficient refrigerant in equipment 2.Charge line problem.	1. Fill equipment tank with more refrigerant. 2..Check charge line, including tank red valve, tank red hose, solenoid #5, solenoid #9 (high side), solenoid #11 (long side), service hoses and HP/LP quick couplers.

During recovery, vacuum pump is pressurized. After period too much oil in vacuum pump	The contact between solenoid valve #8 and valve base is not well sealed.	 Remove solenoid #8 from valve base, clean the solenoid valve and valve base.
During vacuum, there is suction in old oil bottle.	The contact between solenoid valve #2 and valve base is not well sealed.	 Remove solenoid #2 from valve base, clean the solenoid valve and valve base.

Remarks: Regular maintenance by specialized technicians may largely reduce machine failure.